# Team DOOM: Stage 2 Markdown Content V2.0 Submission Report

## 1. Team, Module & Repository Snapshot

| Field | Details |
|---|---|
| **Team Name** | Team DOOM |
| **Module** | Hardware, Networks & Behaviorist AI (Zero-to-One Beginner Learning System) |
| **Repository URL** | https://github.com/reyvaneh/Doom |
| **Branch** | `main` |
| **Snapshot Date** | November 6, 2026 |
| **Latest Commit ID** | `df4bac411cf20626878704baa8c9621dee2d1d87` (Short ID: `df4bac4`) |
| **Latest Markdown Book Commit** | `7edc4c199874bfc2ba4eba762c396007cded9cc9` (Short ID: `7edc4c1`, "Add complete learning materials to markdown_content") |

### Team Members & Module Roles

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| Name | Student ID | GitHub Handle | Stage 2 Markdown & Resource Role |
|---|---|---|---|
| **Neville Reyvan (黄雄明)** | `1820262215` | `@reyvaneh` | Team Lead, submission packaging, full-book PDF compilation, interactive Study Guide & Test Bank HTML integration |
| **Leoric Lius (刘峻源)** | `1820262180` | `@leoriclius` | Unit A (Hardware) core chapters, gap-topic integration, local SVG diagrams, and Arduino lab sketches |
| **Justin Liem (李恩嘉)** | `1820262176` | `@Justin-Liem` | Unit A (Hardware) beginner prerequisites, component walkthroughs, and breadboard lab guide |
| **Carensia Chin (陈云花)** | `1820262207` | `@carensia` | Unit B (Networks) beginner analogies, local network diagram assets, and section organization |
| **Quang Huy Đỗ (杜光辉)** | `1820262177` | `@QuangHuy260608` | Unit B (Networks) Star vs. Mesh topologies, packet routing traces, and protocol comparisons |
| **James /** 池俊賢 | `1820262284` | `@vendouple` | Unit C (Behaviorist AI) chapters, `tools/build_study_book.py` unified book builder, and 46-item Test Bank (`bank.json`) |

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|---|---|---|---|
| **Devi Aurelia (戴薇)** | `1820262183` | `@yrwkwit` | Project Log & QA coordination, beginner readability checks, and cross-resource alignment |

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## 2. Summary of Stage 2 Revisions (Response to Stage 1 Feedback)

Following Professor Zhao Fengnian's Stage 1 feedback (Project Plan: 17/20; Repository & Markdown V1.0: 28/30), the team carried out five substantive revisions in Markdown Content V2.0:

1. **Unified 16-Chapter Numbered Study Book (`Study Book.md` & `markdown_content/learning-materials.md`):** Consolidated separate branch files (`hardware`, `network`, `ai-behavior`) into a single, sequentially numbered 16-chapter curriculum with 10 reference/glossary appendices via `tools/build_study_book.py`. Chapters 1 to 6 teach the core beginner lessons across all three units; Chapters 7 to 15 provide practical lab skills, circuit builds, network traces, and worked Q-learning examples; Chapter 16 integrates Hardware, Networks, and Behaviorist AI into one connected system design activity.
2. **4-Step Beginner Scaffolding (Zero Prerequisite Assumption):** Addressed the feedback that beginners cannot build a full system without learning the basics first. Every unit now progresses from (1) everyday analogies (kitchen prep counter vs. freezer for RAM vs. storage; postal mail for packets) to (2) core definitions and first-principles terms, (3) guided single-step traces and simulations, and (4) system-level integration.
3. **100% Local Image & Asset Storage:** Eliminated external web image dependencies across all Markdown files. All diagrams and photos are stored locally in repository folders (`Learning Materials/Hardware/images/*.svg`, `Learning Materials/Hardware/*.jpg`, `Learning Materials/Network/computernetwork*.jpg`, and `Learning Materials/AIBehaviour/assets/*.svg/.png`), with relative links automatically rebased and verified for both `Learning Materials/` and `markdown_content/`.
4. **Full-Book PDF & Interactive Companions Included:** Added the compiled full-book PDF (`Study Book(Full cover of the markdown content inthe format of books).pdf`), the Stage 2 Study Guide PDF (`DOOM_Stage2_Study_Guide.pdf`), and self-contained interactive HTML files (`DOOM_Stage2_Study_Guide_Interactive.html` and `DOOM_Test_Bank_Interactive.html`) with embedded base64 visuals.
5. **Executable Verification & Practical Supplements:** Added 5 progressive Arduino sketches (`01_light_indicator_basic` through `05_button_override`) for the automatic light indicator lab and a deterministic Python verification model (`Learning Materials/AIBehaviour/checks/rl_vacuum_model.py`) to verify reinforcement learning and reactive vacuum traces.

## 3. Brief Review & Verification Results

- **Automated Build & Link Verification (`tools/build_study_book.py`):** Verified that both `Learning Materials/Study Book.md` and `markdown_content/learning-materials.md` compile from the same canonical unit sources, preserve fenced code blocks, assign unique heading anchors, and resolve local relative links and image paths.
- **Technical & Activity Trace Checks:**
  - *Unit A (Hardware):* Verified binary conversion (`0101` = 5), basic `AND` gate truth tables, Fetch-Decode-Execute register state transitions (`PC -> IR -> ALU -> ACC`), and the automatic light indicator threshold trace across readings `300`, `700`, and `400` (`OFF -> ON -> OFF`) plus hysteresis anti-flicker behavior.
  - *Unit B (Networks):* Verified packet switching sequence reassembly (`Packet 1, 2, 3`), alternate mesh routing (`A -> B -> C`) when a primary link fails, and the separation between local LAN switching, WAN routing, DNS lookup, and HTTPS payload encryption.
  - *Unit C (Behaviorist AI):* Verified priority ordering in Brooks's Subsumption Architecture (Layer 0 collision/battery survival overriding Layer 1 cleaning and Layer 2 wandering) and tabular Q-learning updates via `rl_vacuum_model.py`.
- **ZIP Extraction & Offline Display Check:** Confirmed that when extracted into a standalone directory outside the Git workspace, Markdown files, PDFs, local `.svg`/`.jpg`/`.png` diagrams, and standalone interactive `.html` files open cleanly offline without broken images.

## 4. Cross-Resource Mapping (Markdown V2.0 -> Study Guide, Test Bank, Slides & Videos)

| Markdown Content V2.0 Source | Book Chapters & Appendices | Study Guide & Interactive Simulators | Test Bank Items (`Test Bank/Questions.md` & HTML) | PPT Slides & Video Tutorials |
|---|---|---|---|---|
| `markdown_content/unit-a-hardware.mdLearning` | **Chapters 1, 2, 3, 7, 8, 9, 10**Appendices | Unit A Chapters 1-3, 7-10 + **Interactive CPU** | **16 Items:**`TF01-TF07` | `DOOM_Stage2_PPT_V1.pptx` (Unit A section)**Video 01:** |

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|---|---|---|---|---|
| `Materials/Hardware/hardware.mdLearning Materials/Hardware/beginner-lab-skills.md` | A, D, G, H | **Fetch-Decode-Execute Stepper** | `MC01-MC07OE01-OE02` | Hardware Foundations |
| `markdown_content/unit-b-networks.mdLearning Materials/Network/network_fundamentals.md` | **Chapters 4, 11, 12**Appendices B, E, J | Unit B Chapters 4, 11, 12 + **Interactive Packet Journey & Link-Failure Tracer** | **15 Items:**`TF08-TF14 MC08-MC14OE03-OE04` | `DOOM_Stage2_PPT_V1.pptx` (Unit B section)**Video 02:** Network Communication |
| `markdown_content/unit-c-behaviorist-ai.mdLearning Materials/AIBehaviour/AIBehaviour.md` | **Chapters 5, 6, 13, 14, 15, 16**Appendices C, F, I | Unit C Chapters 5-6, 13-16 + **Interactive Q-Learning 4x4 Grid Simulator** | **15 Items:**`TF15-TF20 MC15-MC20OE05-OE06` | `DOOM_Stage2_PPT_V1.pptx` (Unit C section)**Video 03:** Behaviorist AI & Connected Systems |

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## 5. Selected GitHub Commit Evidence

| *Commit ID* | *Date* | *Author* | *Summary of Contribution* |
|---|---|---|---|
| *`df4bac4`* | *2026-11-06* | *Neville Reyvan (`@reyvaneh`)* | *Uploaded `DOOM_Stage2_PPT_V1.pptx` presentation deck aligned with the 16-chapter curriculum* |
| *`7edc4c1`* | *2026-11-05* | *James (`@vendouple`)* | *Added complete generated learning materials (`learning-materials.md`) and README index to `markdown_content/` with rebased asset links* |
| *`6652701`* | *2026-11-05* | *Neville Reyvan (`@reyvaneh`)* | *Added `Study Book(Full cover of the markdown content inthe format of books).pdf` capturing the full Markdown curriculum in one PDF* |
| *`99f4533` & `1b05d89`* | *2026-11-05* | *Neville Reyvan (`@reyvaneh`)* | *Added and updated `DOOM_Stage2_Study_Guide.pdf` and standalone `DOOM_Stage2_Study_Guide_Interactive.html`* |
| *`9520504` & `8418b7a`* | *2026-11-04* | *James (`@vendouple`)* | *Consolidated unit sources into the 16-chapter numbered `Study Book.md`, polished references, and preserved interactive practice sections* |
| *`3c0bd43`* | *2026-11-04* | *Neville Reyvan (`@reyvaneh`)* | *Created `DOOM_Test_Bank_Interactive.html` (46 auto-graded/guided questions across Hardware, Networks, and AI)* |
| *`594745c`* | *2026-11-04* | *Justin Liem (`@Justin-Liem`)* | *Added hardware unit updates and supporting component materials* |
| *`67a2c49`* | *2026-10-30* | *James (`@vendouple`)* | *Added Stage 2 46-question Test Bank (`Questions.md`, `AnswerKey.md`, `Review.md`, `Doom_Stage2_TestBank_V1.docx`)* |

*(Note: Detailed AI prompts, verification steps, and individual reflections are recorded with side-by-side evidence in the Stage 2 Project Log and individual Personal Logs, kept separate from the public repository per course rules.)*

## 6. Remaining Gaps & Stage 3 Next Steps

1. **Physical Hardware Bench Measurements:** The 5 Arduino light-indicator sketches (`Learning Materials/Hardware/arduino/`) have been verified by code inspection and mathematical trace; recording live multimeter/serial-monitor readings on a physical Arduino Uno board is scheduled for Stage 3 demonstration prep.
2. **Non-Author Beginner Trial Log:** While local technical checks and cross-unit consistency audits are complete, the team will conduct an additional timed trial with a beginner peer during Stage 3 mini-lecture rehearsal to log any remaining questions on Q-table updates and subnet routing.
3. **Stage 3 Mini-Lecture Integration:** Transition the interactive HTML simulators (CPU Fetch-Decode-Execute Stepper, Packet Link-Failure Tracer, and Q-Learning Grid) into live audience prediction polls during the classroom mini-lecture.